

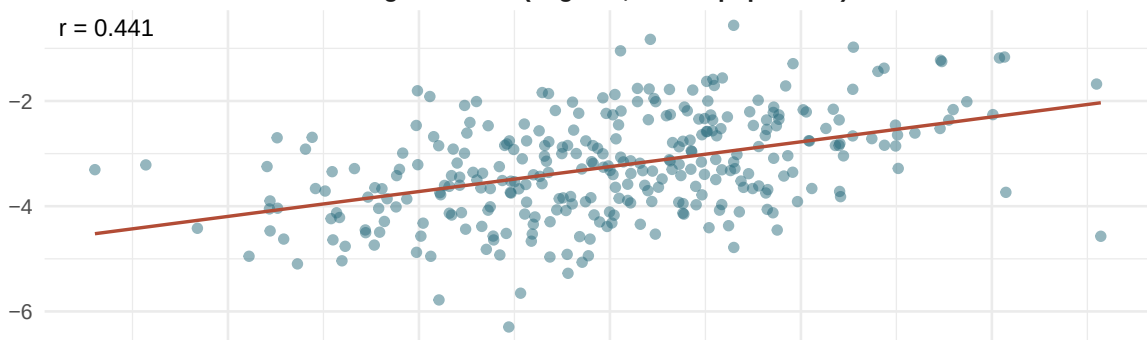
A. Oracle validation: predicted vs. TRUE breeding value

Noise ceiling (observed composite vs. true value): $r = 0.668$

Stage 1: MGIDI (negated, whole-population)

$r = 0.441$

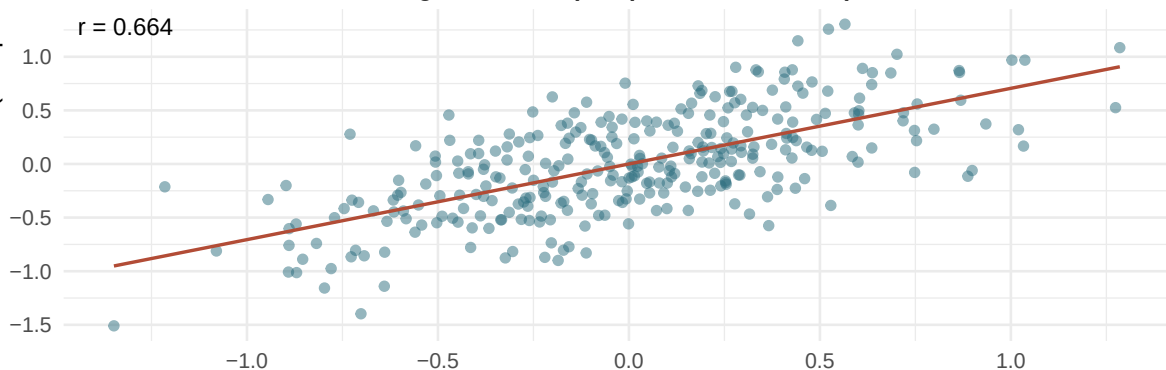
Predicted merit (composite GEBV or negated MGIDI)



Stage 3: GBLUP (composite, out-of-fold)

$r = 0.664$

TRUE composite breeding value (simulation ground truth)



B. Table 5's literal operational gate: NO-GO

95% CI: $[-0.180, 0.022]$; MGIDI not held out here (see README)

